Supplementary material on: Pre-compensation strategies for pressure-driven microfluidic flow in elastic-walled channels (Ströhle and Petit (2026))

Timo Ströhle [**] **Nicolas Petit** [*]

[*] *Centre Automatique et Systèmes (CAS), Mines Paris, Université PSL, 60 bd St Michel, 75006 Paris, France (e-mail: nicolas.petit@minesparis.psl.eu).*
[**] *Institut für techn. Mechanik (ITM), Karlsruhe Institut für Technologie, Kaiserstr. 12, 76131 Karlsruhe, Germany (e-mail: timo.stroehle@kit.edu).*

Abstract: These is supplementary material corresponding to (Ströhle and Petit (2026)), which studies a fundamental problem in pressure-driven fluid mechanics, namely the flow of incompressible fluids within complaint vessels. The compliance of the walls leads to complex fluid–structure interactions, which are well known to practitioners in microfluidics.

1. REACHING A STEADY OUTPUT IN FINITE TIME

The transfer function of interest, that has been derived thoroughly in Ströhle and Petit (2026), is given by

$$G(p) = \exp\left(-\frac{\eta_1 p - \eta_4}{2\eta_2} L\right) \frac{\eta_2}{p - \eta_3} F(p) \tag{1}$$

with

$$F(p) = \frac{\exp(\frac{L}{2}\sqrt{\Delta(p)})}{1 - \exp(L\sqrt{\Delta(p)})} \sqrt{\Delta(p)} \; . \tag{2}$$

This transfer function consists of a delay operator, a linear filter, and $F(p)$ a non rational transfer function. Interestingly, $F(p)$ can be written under a form showing that the singularities associated to $\sqrt{\Delta(p)} = 0$ are reducible. With $d = \eta_1^2 - 4\eta_2$, note

$$\begin{aligned} p_{1,2} &= \left(\eta_1\eta_4 - 2\eta_2\eta_3 \pm 2\sqrt{\eta_2^2\eta_3^2 - \eta_1\eta_2\eta_3\eta_4 + \eta_2\eta_4^2}\right)/d \\ &= \alpha_2 \pm \sqrt{\alpha_3\alpha_1^{-1}} = \alpha_2 \pm \gamma \end{aligned}$$

the two real [1] roots of $\Delta(p) = (p-p_1)(p-p_2)\alpha_1$, with $\alpha_1 \triangleq (\frac{\eta_1}{\eta_2})^2 - \frac{4}{\eta_2}$, $\alpha_2 \triangleq \frac{\eta_1\eta_4/\eta_2^2 - 2\eta_3/\eta_2}{\alpha_1}$, $\alpha_3 \triangleq \alpha_1\alpha_2^2 - (\eta_4/\eta_2)^2$, $\gamma \triangleq \sqrt{\alpha_3\alpha_1^{-1}}$, s.t. $\Delta(p) = \alpha_1((p-\alpha_2)^2 - \gamma^2)$. Using (Abramowitz and Stegun, 1972, 29.2.14 and 29.3.91), a few lines of calculus yield

$$\begin{aligned} \mathcal{L}^{-1}\{(-1/(2F(p))\}(t) &= \mathcal{L}^{-1}\left\{\frac{\sinh(\frac{\sqrt{\Delta}}{2}L)}{\sqrt{\Delta}}\right\}(t) \\ &= \exp(\alpha_2 t)\, I_0\left(\gamma\sqrt{t^2 - \frac{L^2\alpha_1}{4}}\right) \times \mathbb{1}_{\left[-\frac{L\sqrt{\alpha_1}}{2}, \frac{L\sqrt{\alpha_1}}{2}\right]}(t) \; , \end{aligned} \tag{3}$$

where $\mathbb{1}$ is the indicator function and I_0 is the modified Bessel function of the first kind of order 0. Eq. (3) shows that the inverse Laplace transform of $1/F(p)$ has finite support. Therefore, $1/F(p)$ is a finite memory convolution operator. Now $G(p)/F(p)$ is the product of a holomorphic function and a meromorphic function having an isolated point as singularity (a pole at $p = \eta_3$ with multiplicity one). We propose a "poles only" formula for pre-compensation. Consider the input signal defined by its Laplace transform

$$\hat{u}(p) = \frac{1}{p}\frac{1}{F(p)}\left(\frac{\exp(-\eta_3/q) - \exp(-p/q)}{\exp(-\eta_3/q) - 1}\right)^m \frac{1}{G(0)} \tag{4}$$

with m a positive natural number, and q a positive real number. The signal (4) cancels the singularity of $G(s)$. Indeed, $G(p)\hat{u}(p)$ is an *entire* function (globally holomorphic on the whole complex plane $\mathbb{C}$). It is also of *exponential type*, i.e. there exist A and C s.t. $|G(p)u(p)| \leq Ce^{A|p|}$. Then, using the Paley-Wiener theorem (Rudin, 1987, page 375), its inverse Laplace transform is a finite support function. Denote its support by $[0, T]$. The expansion of $\hat{u}(p)$ involves a finite number of delay operators and the inverse $1/F(p)$ which we interpreted as another finite support operator. The value of T is bounded by $m/q + \sqrt{\alpha_1}$. The signal is the convolution of a sequence of $m+1$ steps, convoluted by the finite memory filter $1/F(p)$. This signal provides a unitary output response in the time T, as shown by a direct application of the final value theorem, see e.g. Churchill (1972). Illustrations are given in Fig. 1, with

[1] As the stiffness becomes large enough (Assumption 1 in Ströhle and Petit (2026)), η_2 grows (in absolute value), while η_1, η_3 and η_4 converge to finite values. Therefore the $\eta_2^2\eta_3^2 > 0$ becomes dominant in the expression $\eta_2^2\eta_3^2 - \eta_1\eta_2\eta_3\eta_4 + \eta_2\eta_4^2$ and the roots are real.

$q = 1/2$, $m = 2$, $T \approx 4.50$, which can be compared with Fig. 2 in Ströhle and Petit (2026)

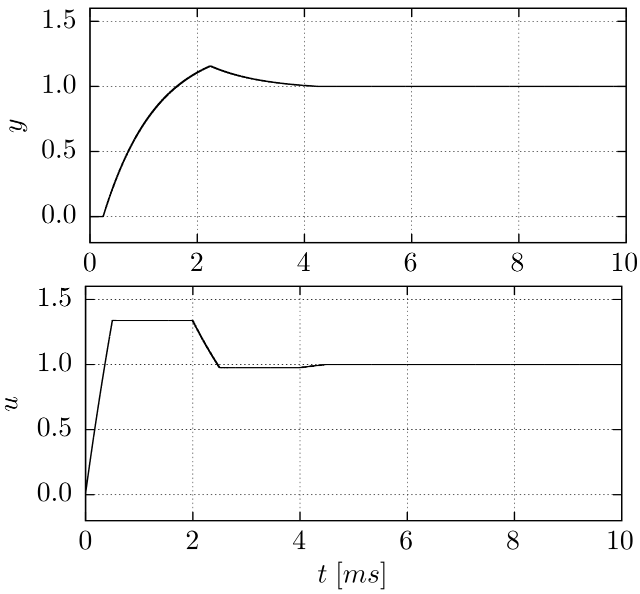

Fig. 1. Finite time unitary transient for the output (top) using a simple input signal (bottom).

2. ANOTHER VARIABLE OF INTEREST: INPUT FLOWRATE RESPONSE TO INPUT PRESSURE

Following Eq. (9) in Ströhle and Petit (2026), the inlet-flowrate takes the form $\hat{b}(0,p) = l_1(p) + l_2(p)$, and $H(p) \triangleq \hat{b}(0,p)/\hat{u}(p)$ is given by

$$\frac{1}{R_1R_2}\frac{1}{(1-\exp((R_2-R_1)L))}(R_2\exp((R_2-R_1)L) - R_1)p\ . \tag{5}$$

After a few lines of calculus, using $(1+\exp(+L\sqrt{\Delta}))(1-\exp(+L\sqrt{\Delta}))^{-1} = 2\cdot(1-\exp(L\sqrt{\Delta}))^{-1}-1$, we get

$$\begin{aligned}H(p) &= \frac{1}{2}\left(\frac{\eta_1 p-\eta_4}{p-\eta_3} + \frac{\eta_2}{(p-\eta_3)}\frac{1+\exp(+L\sqrt{\Delta})}{1-\exp(+L\sqrt{\Delta})}\sqrt{\Delta}\right)\\ &= \frac{1}{2}\left(\frac{\eta_1 p-\eta_4}{p-\eta_3} + \frac{\eta_2}{p-\eta_3}\frac{2\sqrt{\Delta}}{1-\exp(L\sqrt{\Delta})} - \frac{\eta_2}{p-\eta_3}\sqrt{\Delta}\right).\end{aligned}$$

One easily verifies that $H(0) = G(0)$. We now determine the step response by computing $\mathcal{L}^{-1}\left\{\frac{1}{p}H(p)\right\}$ as the sum of three terms $T_1 + T_2 + T_3$.

First term. The first term gives

$$\frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{p}\frac{\eta_1 p-\eta_4}{p-\eta_3}\right\} = \frac{1}{2}\left(\left(\eta_1 - \frac{\eta_4}{\eta_3}\right)\exp(\eta_3 t) + \frac{\eta_4}{\eta_3}\right) \triangleq T_1.$$

Second term. For the second term, the inverse Laplace transform is given by a convolution of the inverse Laplace transform of

$$\mathcal{L}^{-1}\left\{\frac{1}{p}\frac{\eta_2}{p-\eta_3}\right\} = \frac{\eta_2}{\eta_3}(\exp(\eta_3 t) - 1) \triangleq \chi(t)$$

and the function $\sqrt{\Delta}/(1-\exp(L\sqrt{\Delta}))$, which can be derived analogously to Sec. 3.3 in Ströhle and Petit (2026) by defining the coefficients $\nu_k = L\alpha_1^{\frac{1}{2}}(k+1)$, as follows

$$\begin{aligned}&\mathcal{L}^{-1}\left\{\frac{\sqrt{\Delta}}{1-\exp(L\sqrt{\Delta})}\right\}(t)\\ &= \alpha_1^{\frac{1}{2}}\sum_{k=0}^{\infty}\left[\left(\gamma^2\Theta(t-\nu_k) - \gamma^2\nu_k\delta(t-\nu_k)\right)\frac{1}{z}I_1(z)\right.\\ &\left.- \left(\frac{\gamma^4\nu_k^2}{z^2}I_2(z) + \delta'(t-\nu_k)\right)\Theta(t-\nu_k)\right]\exp(\alpha_2 t) \triangleq \kappa(t)\ .\end{aligned}$$

where we redefine $z \triangleq \gamma\sqrt{t^2-\nu_k^2}$. Hence,

$$\mathcal{L}^{-1}\left\{\frac{1}{p}\frac{\eta_2}{p-\eta_3}\frac{\sqrt{\Delta}}{1-\exp(L\sqrt{\Delta})}\right\}(t) = \int_0^{\infty}\chi(\tau)\cdot\kappa(t-\tau)\,d\tau.$$

Using of the identities

$$\left.\begin{aligned}(f\cdot\delta_a) * g(t) &= f(a)g(t-a)\\ (f\cdot\Theta_a) * g(t) &= \int_a^{\infty} f(s)g(t-s)ds\\ (f\cdot\Theta_a\cdot\delta_a') * g(t) &= f(a)g'(t-a) - f'(a)g(t-a)\end{aligned}\right\} \tag{6}$$

and defining $\sigma(t) \triangleq \alpha_1^{\frac{1}{2}}\exp(\alpha_2 t)$ we get

$$\begin{aligned}T_2 &\triangleq \mathcal{L}^{-1}\left\{\frac{1}{p}\frac{\eta_2}{p-\eta_3}\frac{\sqrt{\Delta}}{1-\exp(L\sqrt{\Delta})}\right\}(t)\\ &= \sum_{k=0}^{\infty}\left[\int_{\nu_k}^{\infty}\left(I_1(z(\tau)) - \nu_k^2\gamma^2\frac{I_2(z(\tau))}{z(\tau)}\right)\frac{\gamma^2\sigma(\tau)}{z(\tau)}\cdot\chi(t-\tau)\,d\tau\right.\\ &\qquad\left.-\sigma(\nu_k)\left(\left(\frac{\gamma^2\nu_k}{2}-\alpha_2\right)\cdot\chi(t-\nu_k) + \chi'(t-\nu_k)\right)\right]\end{aligned}$$

Third term. Since

$$\begin{aligned}\frac{1}{p}\sqrt{\Delta} &= \alpha_1^{\frac{1}{2}}\frac{(p-\alpha_2)^2-\gamma^2}{p}\frac{1}{\sqrt{(p-\alpha_2)^2-\gamma^2}}\\ &= \left(p - 2\alpha_2 + \frac{1}{p}(\alpha_2^2-\gamma^2)\right)\frac{\alpha_1^{\frac{1}{2}}}{\sqrt{(p-\alpha_2)^2-\gamma^2}}\ ,\end{aligned}$$

the third term, T_3, is itself given by the convolution of three functions.

First, $\frac{1}{2}\mathcal{L}^{-1}\{\eta_2/(p-\eta_3)\}(t) = \eta_2\exp(\eta_3 t)/2\,\Theta(t) \triangleq \kappa_2(t)$ and $\mathcal{L}^{-1}\{p - 2\alpha_2 + (\alpha_2^2-\gamma^2)/p\}(t) = \delta'(t) - 2\alpha_2\delta(t) + \alpha_2^2 - \gamma^2 \triangleq \kappa_3(t)$, the convolution product of being, accounting for the initial jump of κ_2, s.t. $(\kappa_2 * \kappa_3)(t) = \left(\frac{1}{2}\eta_2\eta_3 - \alpha_2\eta_2\right)\exp(\eta_3 t) - \frac{1}{2}(\alpha_2^2-\gamma^2)\frac{\eta_2}{\eta_3} + \frac{1}{2}\eta_2\delta(t)$ and finally $\mathcal{L}^{-1}\left\{\frac{\alpha_1^{\frac{1}{2}}}{\sqrt{(p-\alpha_2)^2-\gamma^2}}\right\}(t) = \alpha_1^{\frac{1}{2}}I_0(\gamma\cdot t)\cdot\exp(\alpha_2 t) = \sigma(t)I_0(\gamma\cdot t)$. (cf. (Abramowitz and Stegun, 1972, Eq. 29.3.59/Eq. 29.3.60, p. 1025))

Thus, the response of the inlet flowrate, denoted by $b(0,t)$, to a unit step input $u(t)$ is given by

$$\begin{aligned}b(0,t) &= \frac{1}{2}\left(\left(\eta_1 - \frac{\eta_4}{\eta_3}\right)\exp(\eta_3 t) + \frac{\eta_4}{\eta_3}\right)\\ &+ \sum_{k=0}^{\infty}\left[\int_{\nu_k}^{\infty}\left(\frac{I_1(z(\tau))}{z(\tau)} - \nu_k^2\gamma^2\frac{I_2(z(\tau)}{z^2(\tau)}\right)\gamma^2\sigma(\tau)\cdot\chi(t-\tau)\,d\tau\right.\\ &\qquad\left.-\left(\left(\frac{\gamma^2\nu_k}{2}-\alpha_2\right)\cdot\chi(t-\nu_k) + \chi'(t-\nu_k)\right)\sigma(\nu_k)\right]\\ &- \int_0^{\infty}\sigma(\tau)I_0(\gamma\cdot\tau)\cdot\kappa_4(t-\tau)\,d\tau - \frac{1}{2}\eta_2\sigma(t)I_0(\gamma\cdot t)\ ,\end{aligned} \tag{7}$$

where $\kappa_4 \triangleq \left(\frac{1}{2}\eta_2\eta_3 - \alpha_2\eta_2\right)\exp(\eta_3 t) - \frac{1}{2}(\alpha_2^2-\gamma^2)\frac{\eta_2}{\eta_3}$. Fig. 2 reports the comparison of the inverse Laplace transform of

$H(p)/p$, derived analytically in this section and presented in (7) with the results of a numerical computation of the inverse Laplace transform of the step-response using the algorithm presented in Ströhle and Petit (2025). All functions are normalized with respect to the given static gain

$$y_\infty \triangleq \left(\frac{\hat{y}}{\hat{u}}\right)(p=0) = \frac{\eta_4}{\eta_3 - \eta_3 \exp(-\frac{\eta_4}{\eta_2}L)} \,. \tag{8}$$

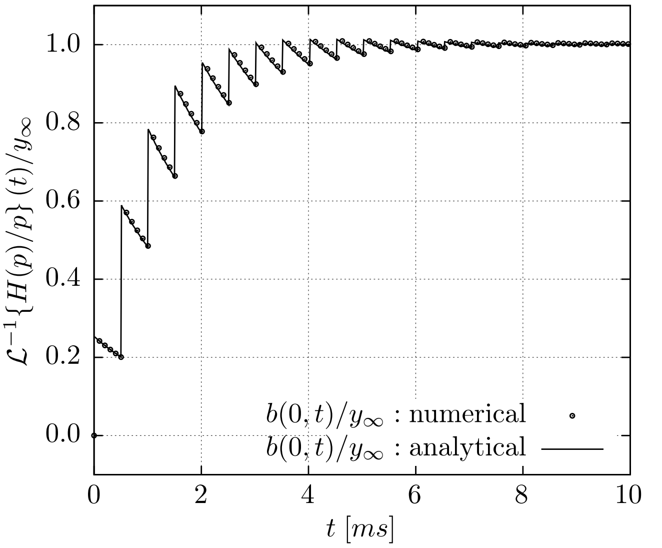

Fig. 2. Input flowrate response to a step in input pressure. Comparison of an analytically derived inverse Laplace transform of the input pressure to input flowrate step response (7) and a numerically computed step-response obtained by the algorithm presented in Ströhle and Petit (2025).

3. NUMERICAL VERIFICATION OF ASM. 1 IN Ströhle and Petit (2026)

The steady state $\tilde{z} = (\tilde{Q}, \tilde{A})$ of the problem at hand is given by

$$g(\tilde{Q}, \tilde{A})\, \frac{d\tilde{A}}{ds} = -K\frac{\tilde{Q}}{\tilde{A}} - \frac{\tilde{A}}{\rho} f(\tilde{A}) \frac{d\beta}{ds} \,.$$

This ordinary differential equation (ODE) is readily solved numerically (backwards, staying always from the singularity $A^{\text{eq}} = 0$) from the boundary condition at $s = 1$ (cf. Ströhle and Petit (2025)). The partial differential equation at hand, linearized around this steady state is given by

$$\partial_t a + \partial_s b = 0 \,,\ \partial_t b + \eta_1 \partial_s b - \eta_2 \partial_s a = \eta_3 b - \eta_4 a \tag{9}$$

Herein, the space-varying coefficients η_i tend to become constant when increasing the stiffness $\beta(s)$ uniformly.

In Fig. 3 the numerically computed space-varying coefficients $\eta_i(s)$ are depicted for increasing $\beta_j(s)$, i.e. $\eta_i(s; \beta_j(s))$, with

$$\beta_j(s) = \frac{1}{j+1}\beta_0(s) + j \qquad \forall j \in \{0, 2, 5, 10, 100\}.$$

Herein the function $\beta_0(s)$ is taken from Ströhle and Petit (2025).

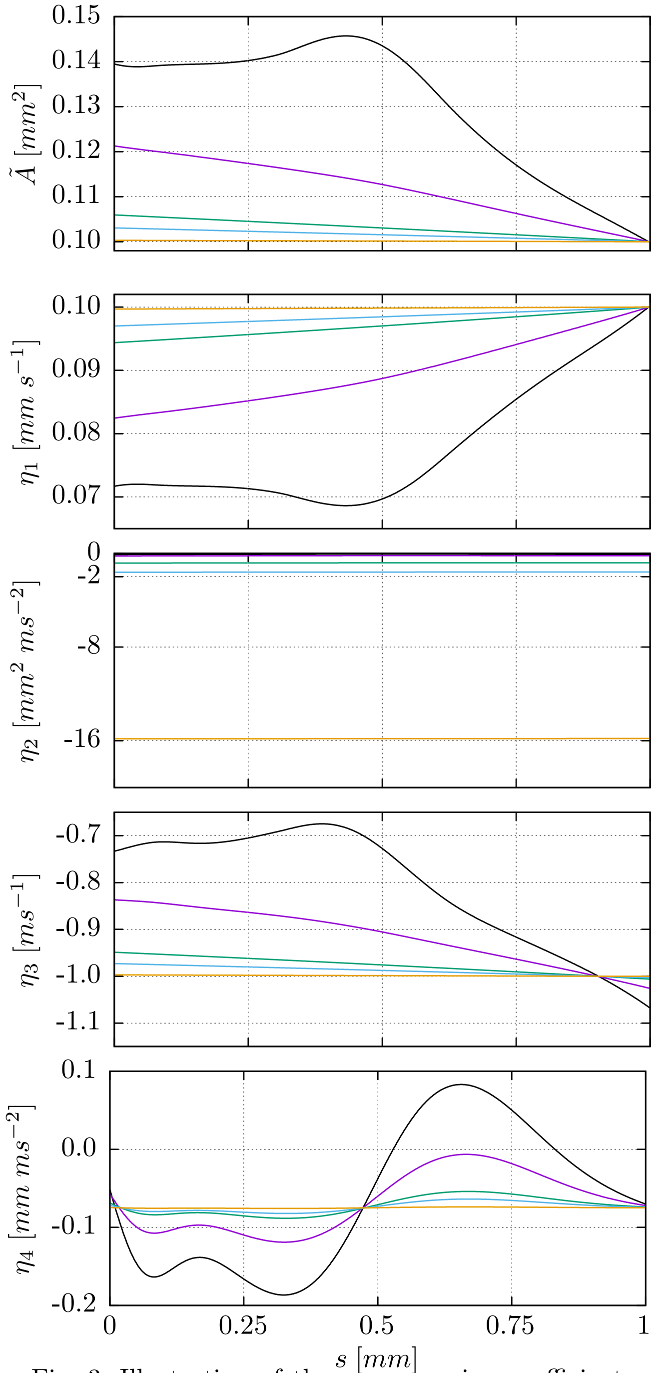

Fig. 3. Illustration of the space-varying coefficients $\eta_i(s)$ for increasing stiffness $\beta(s)$.

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